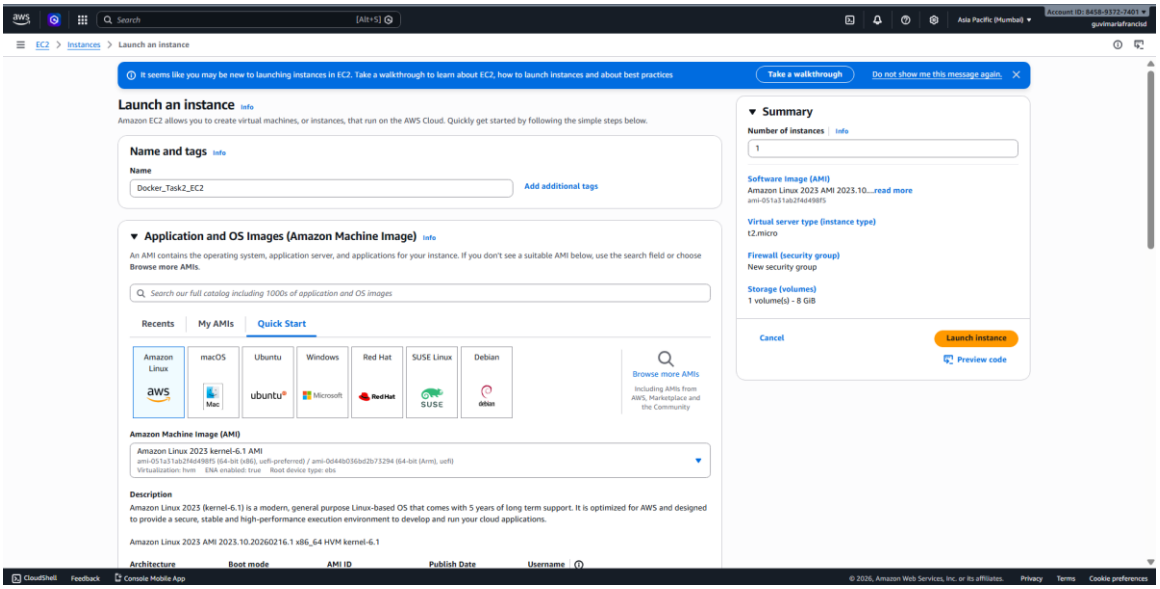
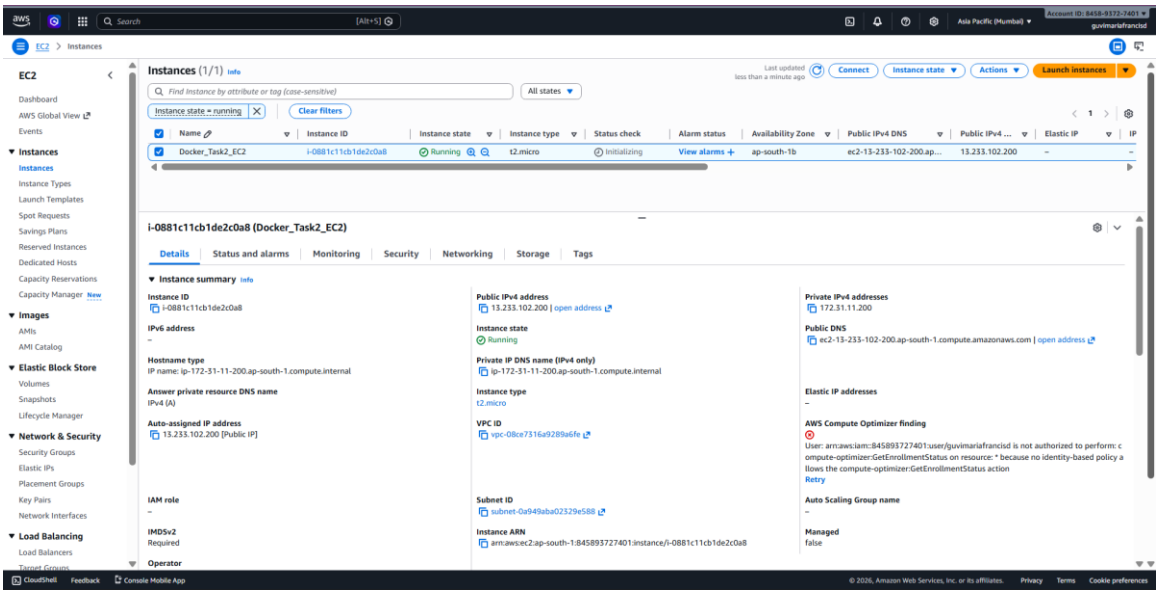
Docker Task -2

Task Description:

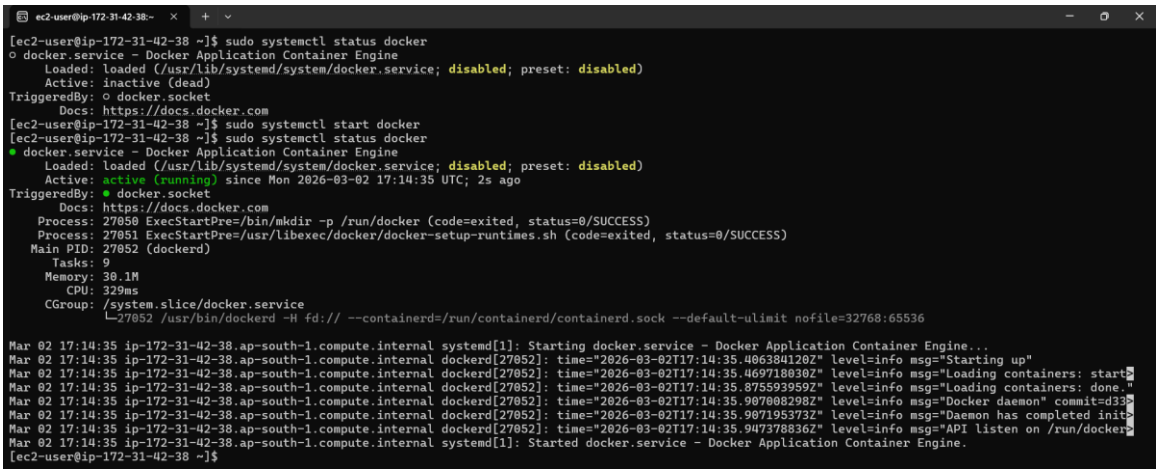
Create a Docker file, docker-compose file which when executed must display your basic details in the website.

1. Creating an EC2 machine:





2. Initiating Docker:



```
[ec2-user@ip-172-31-42-38 ~]$ sudo systemctl status docker
○ docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: disabled)
     Active: inactive (dead)
TriggeredBy: ○ docker.socket
       Docs: https://docs.docker.com
[ec2-user@ip-172-31-42-38 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-42-38 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: disabled)
     Active: active (running) since Mon 2026-03-02 17:14:35 UTC; 2s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
    Process: 27050 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
    Process: 27051 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
   Main PID: 27052 (dockerd)
      Tasks: 9
     Memory: 30.1M
        CPU: 329ms
     CGroup: /system.slice/docker.service
             └─27052 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.4063841207" level=info msg="Starting up"
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.4697180307" level=info msg="Loading containers: start
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.8755939597" level=info msg="Loading containers: done."
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.9070082987" level=info msg="Docker daemon" commit=d33
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.9071953737" level=info msg="Daemon has completed init
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.9473788367" level=info msg="API listen on /run/docker
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal systemd[1]: Started docker.service - Docker Application Container Engine.
[ec2-user@ip-172-31-42-38 ~]$
```

```
ec2-user@ip-172-31-42-38:~
[ec2-user@ip-172-31-42-38 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-42-38 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-03-02 17:14:35 UTC; 1min 17s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 27052 (dockerd)
      Tasks: 9
     Memory: 30.1M
        CPU: 333ms
     CGroup: /system.slice/docker.service
             └─27052 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.406384120Z" level=info msg="Starting up"
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.469718030Z" level=info msg="Loading containers: start
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.875593959Z" level=info msg="Loading containers: done."
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.907008298Z" level=info msg="Docker daemon" commit=d33
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.907195373Z" level=info msg="Daemon has completed init
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal dockerd[27052]: time="2026-03-02T17:14:35.947378836Z" level=info msg="API listen on /run/docker
Mar 02 17:14:35 ip-172-31-42-38.ap-south-1.compute.internal systemd[1]: Started docker.service - Docker Application Container Engine.
[ec2-user@ip-172-31-42-38 ~]$
```

3. Creating directory and files for the project:

```
ec2-user@ip-172-31-42-38:~/
[ec2-user@ip-172-31-42-38 ~]$ mkdir docker_task2
[ec2-user@ip-172-31-42-38 ~]$ cd docker_task2/
[ec2-user@ip-172-31-42-38 docker_task2]$ ls
[ec2-user@ip-172-31-42-38 docker_task2]$ pwd
/home/ec2-user/docker_task2
[ec2-user@ip-172-31-42-38 docker_task2]$ vi index.html
[ec2-user@ip-172-31-42-38 docker_task2]$ cat index.html
<!DOCTYPE html>
<html>
<head>
<title>DevOps Docker Project</title>
</head>

<body style="font-family: Arial; text-align:center; margin-top:100px;">

<h1>Hello from Docker 🚀</h1>

<h2>Basic Details</h2>

<p><b>Name:</b> Francis</p>
<p><b>Role:</b> DevOps Engineer</p>
<p><b>Project:</b> Docker Website on AWS EC2</p>

</body>
</html>
[ec2-user@ip-172-31-42-38 docker_task2]$
```

```
ec2-user@ip-172-31-42-38:~/
<!DOCTYPE html>
<html>
<head>
<title>DevOps Docker Project</title>
</head>

<body style="font-family: Arial; text-align:center; margin-top:100px;">

<h1>Hello from Docker 🚀</h1>

<h2>Basic Details</h2>

<p><b>Name:</b> Francis</p>
<p><b>Role:</b> DevOps Engineer</p>
<p><b>Project:</b> Docker Website on AWS EC2</p>

</body>
</html>
~
```

4. **Creating the Docker File to pull the nginx image and host the index.html file**

```
ec2-user@ip-172-31-42-38:~/

[ec2-user@ip-172-31-42-38 docker_task2]$ ls
Dockerfile  index.html
[ec2-user@ip-172-31-42-38 docker_task2]$ cat Dockerfile
FROM nginx:latest

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80

[ec2-user@ip-172-31-42-38 docker_task2]$
```

```
ec2-user@ip-172-31-42-38:~/

FROM nginx:latest

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80
```

5. **Creating a custom image to use for container creation:**

```
ec2-user@ip-172-31-42-38:~/

[ec2-user@ip-172-31-42-38 docker_task2]$ pwd
/home/ec2-user/docker_task2
[ec2-user@ip-172-31-42-38 docker_task2]$ ls -l
total 8
-rw-r--r--. 1 ec2-user docker  80 Mar  2 17:35 Dockerfile
-rw-r--r--. 1 ec2-user docker 336 Mar  2 17:28 index.html
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$ docker build -t docker_task2_image .
[+] Building 1.0s (7/7) FINISHED                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                          0.0s
 => => transferring dockerfile: 176B                                                                          0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                                               0.9s
 => [internal] load .dockerignore                                                                             0.0s
 => => transferring context: 2B                                                                               0.0s
 => [internal] load build context                                                                             0.0s
 => => transferring context: 91B                                                                              0.0s
 => [1/2] FROM docker.io/library/nginx:latest@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e  0.0s
 => CACHED [2/2] COPY index.html /usr/share/nginx/html/index.html                                             0.0s
 => exporting to image                                                                                        0.0s
 => => exporting layers                                                                                       0.0s
 => => writing image sha256:dcda30ca5032c758c6f22983e87855cc1785b998b5eab1f4bdc1279a2b32e08c                  0.0s
 => => naming to docker.io/library/docker_task2_image                                                         0.0s
[ec2-user@ip-172-31-42-38 docker_task2]$ docker images
REPOSITORY          TAG       IMAGE ID       CREATED          SIZE
docker_task2_image  latest    dcda30ca5032   5 minutes ago    161MB
hello-world         latest    1b44b5a3e06a   6 months ago     10.1kB
[ec2-user@ip-172-31-42-38 docker_task2]$
```

6. **Creating docker container using the custom created image:**

```
ec2-user@ip-172-31-42-38:~/

[ec2-user@ip-172-31-42-38 docker_task2]$ docker images
REPOSITORY          TAG       IMAGE ID       CREATED          SIZE
docker_task2_image  latest    dcda30ca5032   13 minutes ago   161MB
hello-world         latest    1b44b5a3e06a   6 months ago     10.1kB
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$ docker run -d -p 80:80 --name DT2_C docker_task2_image
c065dd0dcab035654d3775f3723b9051737ce442655864150e26d5d0f277f8e7
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$ docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED         STATUS         PORTS                                   NAMES
c065dd0dcab0   docker_task2_image   "/docker-entrypoint.…"   3 seconds ago   Up 2 seconds   0.0.0.0:80->80/tcp, :::80->80/tcp   DT2_C
[ec2-user@ip-172-31-42-38 docker_task2]$
```

7. **Verifying the status by searching the public IP:**

DevOps Docker Project

Not secure 13.201.9.39

Incognito

Hello from Docker

Basic Details

Name: Francis

Role: DevOps Engineer

Project: Docker Website on AWS EC2

8. Creating a docker compose file:

```
[ec2-user@ip-172-31-42-38 docker_task2]$ docker stop DT2_C
DT2_C
[ec2-user@ip-172-31-42-38 docker_task2]$ docker rm DT2_C
DT2_C
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$ vi docker-compose.yml
```

```
services:
  web:
    build: .
    ports:
      - "80:80"
~
```

9. Creatin the image and container using the Docker Compose file:

```
[ec2-user@ip-172-31-42-38 docker_task2]$ docker images
REPOSITORY          TAG       IMAGE ID       CREATED          SIZE
docker_task2_image  latest    dcda30ca5032   21 minutes ago   161MB
hello-world         latest    1b44b5a3e06a   6 months ago     10.1kB
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$ docker rmi docker_task2_image
Untagged: docker_task2_image:latest
Deleted: sha256:dcda30ca5032c758c6f22983e87855cc1785b998b5eab1f4bdc1279a2b32e08c
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
hello-world   latest    1b44b5a3e06a   6 months ago   10.1kB
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$ docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED          STATUS                     PORTS     NAMES
11e46ba6ba68   hello-world   "/hello"   43 minutes ago   Exited (0) 43 minutes ago            heuristic_solomon
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$
```

10. Composing the docker container using the docker compose file:

```
[ec2-user@ip-172-31-42-38 docker_task2]$ docker-compose up -d
[+] Building 1.5s (9/9) FINISHED
 => [internal] load local bake definitions                                                    0.0s
 => => reading from stdin 504B                                                                0.0s
 => [internal] load build definition from Dockerfile                                          0.0s
 => => transferring dockerfile: 176B                                                          0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                               1.2s
 => [internal] load .dockerignore                                                             0.0s
 => => transferring context: 2B                                                               0.0s
 => [internal] load build context                                                             0.0s
 => => transferring context: 429B                                                             0.0s
 => CACHED [1/2] FROM docker.io/library/nginx:latest@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e   0.0s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html                                    0.0s
 => exporting to image                                                                        0.0s
 => => exporting layers                                                                       0.0s
 => => writing image sha256:166bb9becacb7cf19480c1079b8e158689dfeb151cced155bf85e392f21dd8fc   0.0s
 => => naming to docker.io/library/docker_task2-web                                           0.0s
 => resolving provenance for metadata file                                                    0.0s
[+] up 3/3
 ✔Image docker_task2-web       Built                                                          1.6s
 ✔Network docker_task2_default Created                                                        0.2s
 ✔Container docker_task2-web-1 Started                                                        0.6s
[ec2-user@ip-172-31-42-38 docker_task2]$ docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED          STATUS         PORTS                                   NAMES
bc5caf7857ad   docker_task2-web   "/docker-entrypoint.…"   10 seconds ago   Up 9 seconds   0.0.0.0:80->80/tcp, :::80->80/tcp   docker_task2-web-1
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$ docker images
REPOSITORY         TAG       IMAGE ID       CREATED          SIZE
docker_task2-web   latest    166bb9becacb   14 seconds ago   161MB
hello-world        latest    1b44b5a3e06a   6 months ago     10.1kB
[ec2-user@ip-172-31-42-38 docker_task2]$
[ec2-user@ip-172-31-42-38 docker_task2]$
```

11. Checking the output in public IP:

DevOps Docker Project | DevOps Docker Project

Not secure 13.201.9.39

Hello from Docker

Basic Details

Name: Francis

Role: DevOps Engineer

Project: Docker Website on AWS EC2